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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-
selectedcolumns.csv")

print(df.head())
print(df.shape)
```

```
   Age Attrition     BusinessTravel  DailyRate              Department

    41       Yes      Travel_Rarely       1102                   Sales

\
0

1   49        No  Travel_Frequently        279  Research & Development

2   37       Yes      Travel_Rarely       1373  Research & Development

3   33        No  Travel_Frequently       1392  Research & Development

4   27        No      Travel_Rarely        591  Research & Development


   DistanceFromHome  Education EducationField  EmployeeCount
EmployeeNumbe
                  1          2  Life Sciences              1 0
                                                             1
                  8          1  Life Sciences              1 1
                                                             2
                  2          2          Other              1 2
                                                             4
                  3          4  Life Sciences              1 3
                                                             5
                  2          1        Medical              1 4
                                                             7

(1470, 10)
```

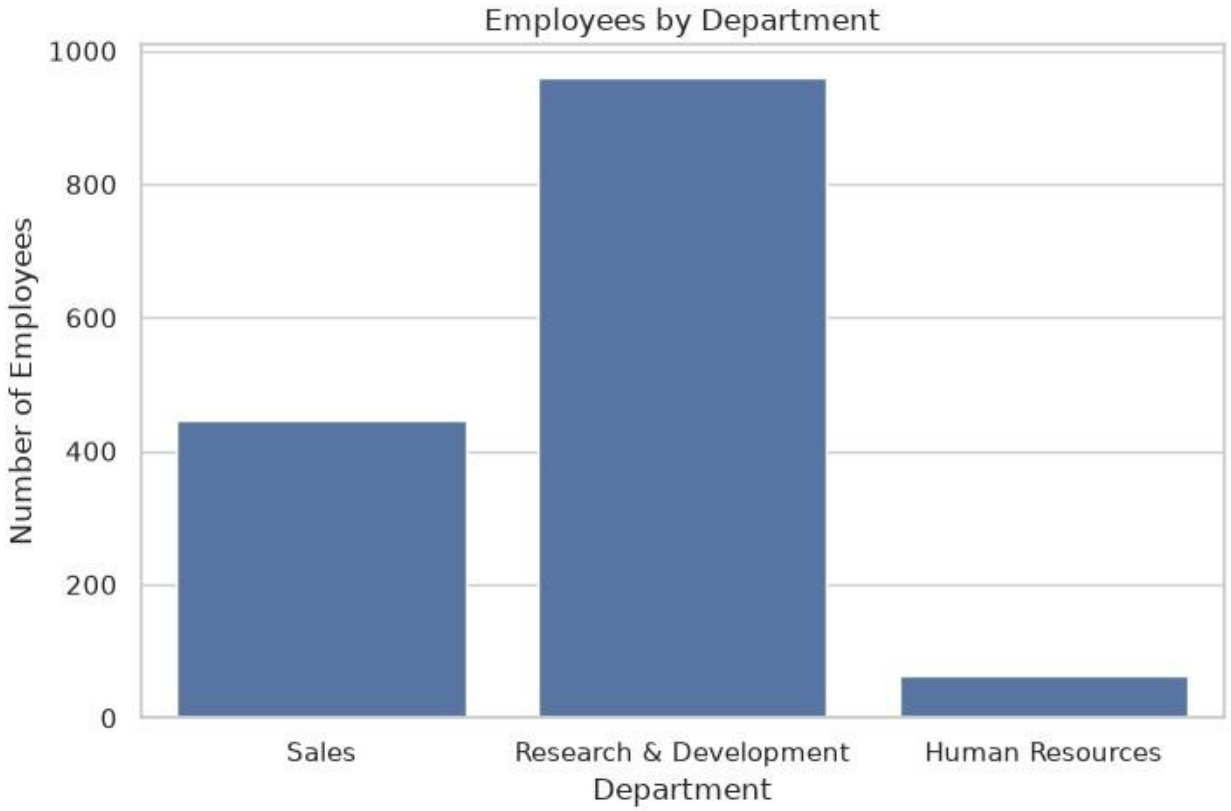
```
print(df.columns.tolist())
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
```

```
'EmployeeNumber']
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x="Department")
```

```
plt.title("Employees by Department")
plt.xlabel("Department") plt.ylabel("Number of
Employees") plt.show()
```



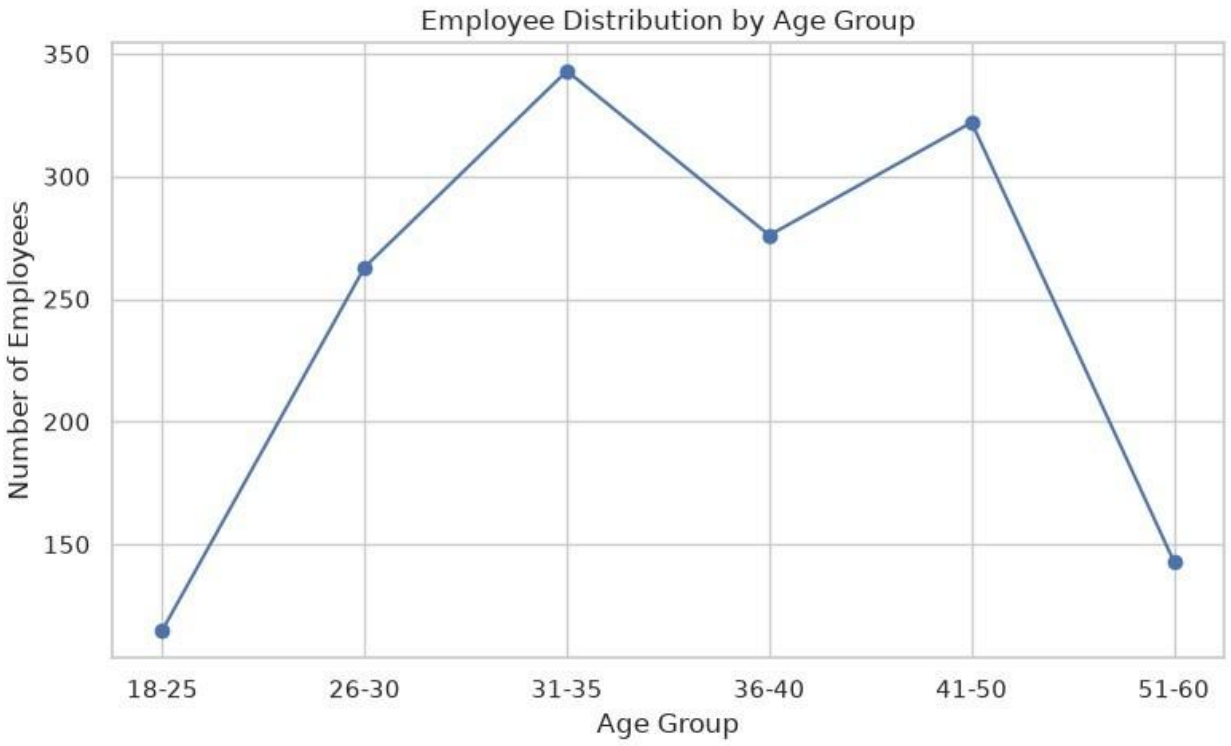

```
df["AgeGroup"] = pd.cut(
df["Age"],
    bins=[18, 25, 30, 35, 40, 50, 60],
    labels=["18-25", "26-30", "31-35", "36-40", "41-50", "51-60"]
)

age_count = df["AgeGroup"].value_counts().sort_index()

plt.figure(figsize=(9, 5))

plt.plot(age_count.index, age_count.values, marker="o")

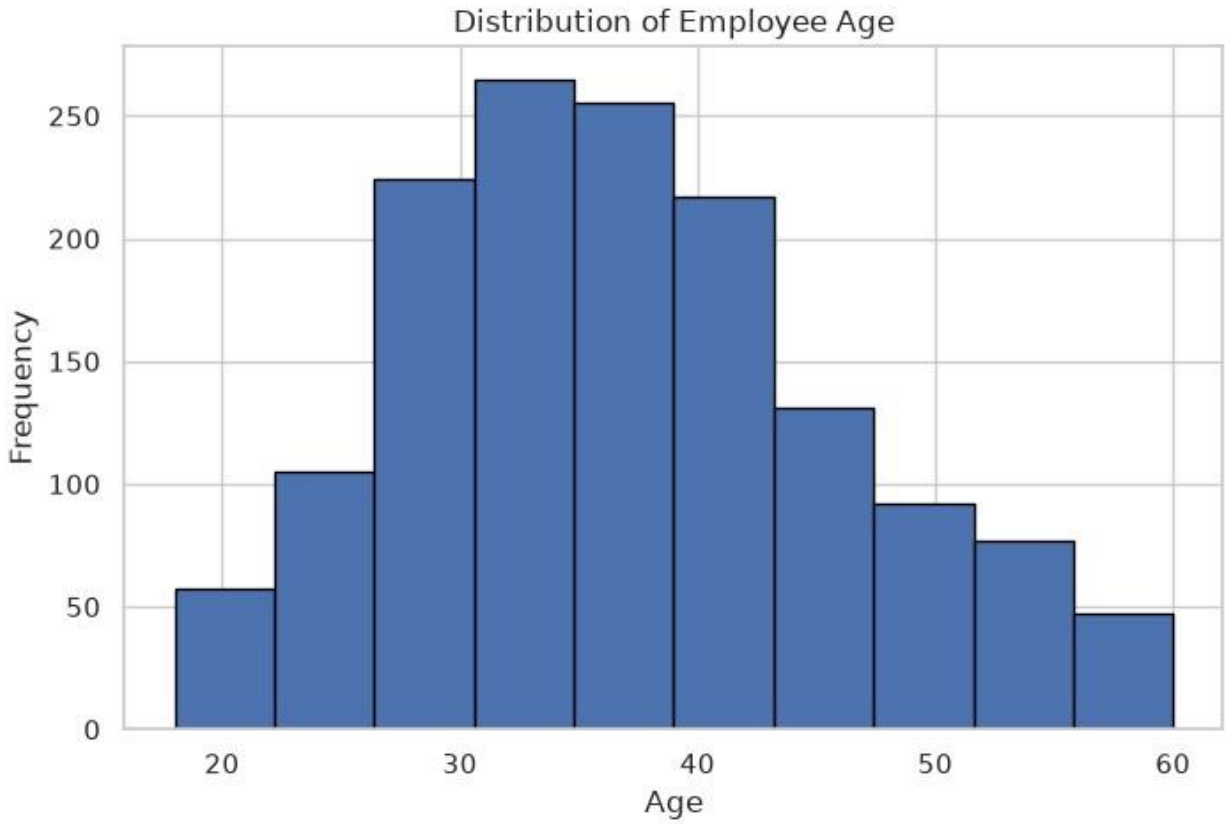
plt.title("Employee Distribution by Age Group")
plt.xlabel("Age Group") plt.ylabel("Number of
Employees")
plt.show()
```




```
plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=10, edgecolor="black")

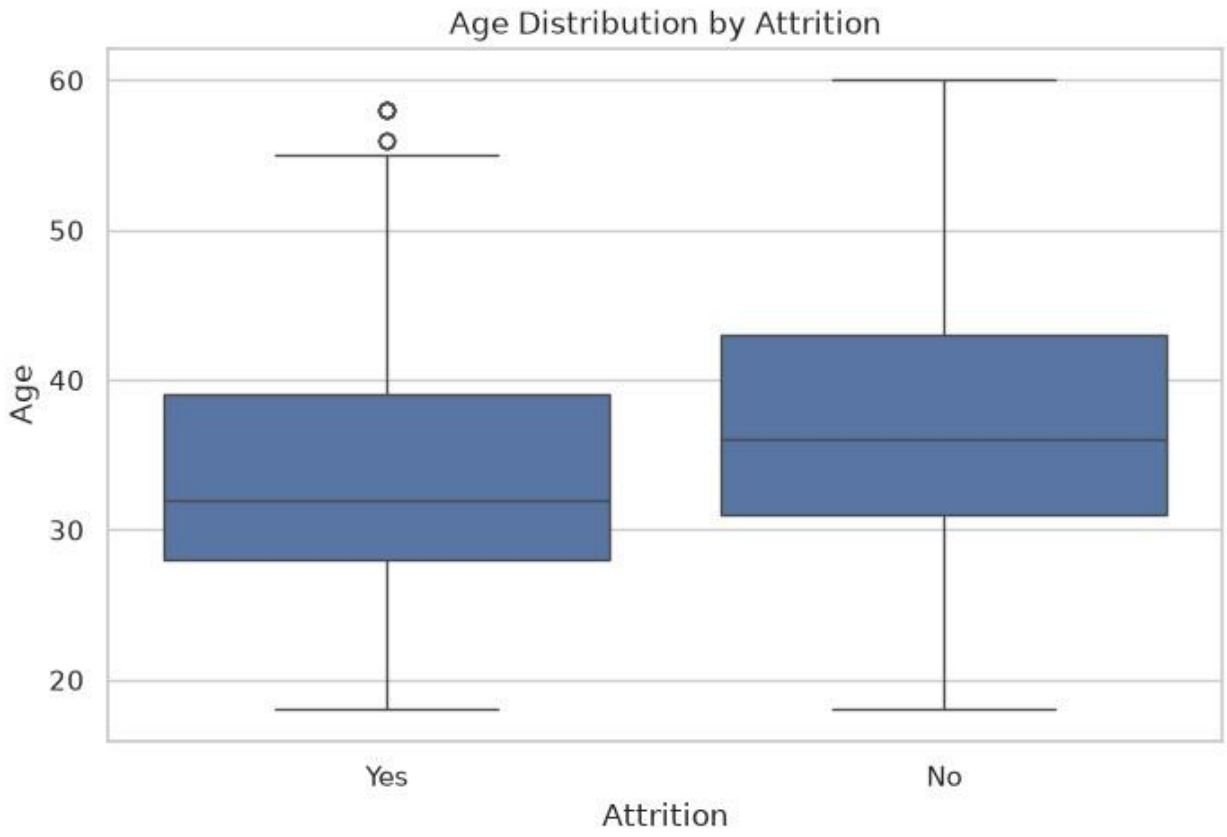
plt.title("Distribution of Employee Age")
plt.xlabel("Age") plt.ylabel("Frequency")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Attrition", y="Age")

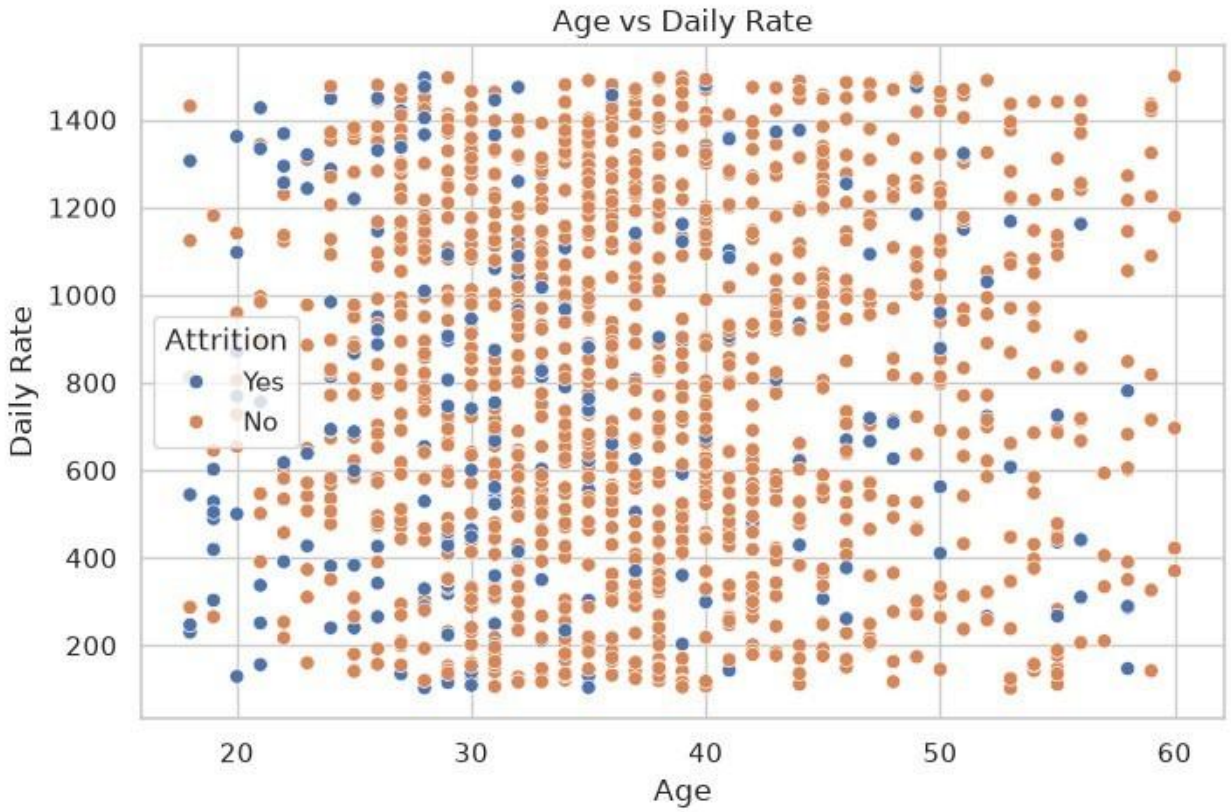
plt.title("Age Distribution by Attrition")
plt.xlabel("Attrition") plt.ylabel("Age")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.scatterplot(
data=df,
x="Age",
y="DailyRate",
hue="Attrition"
)

plt.title("Age vs Daily Rate")
plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.show()
```

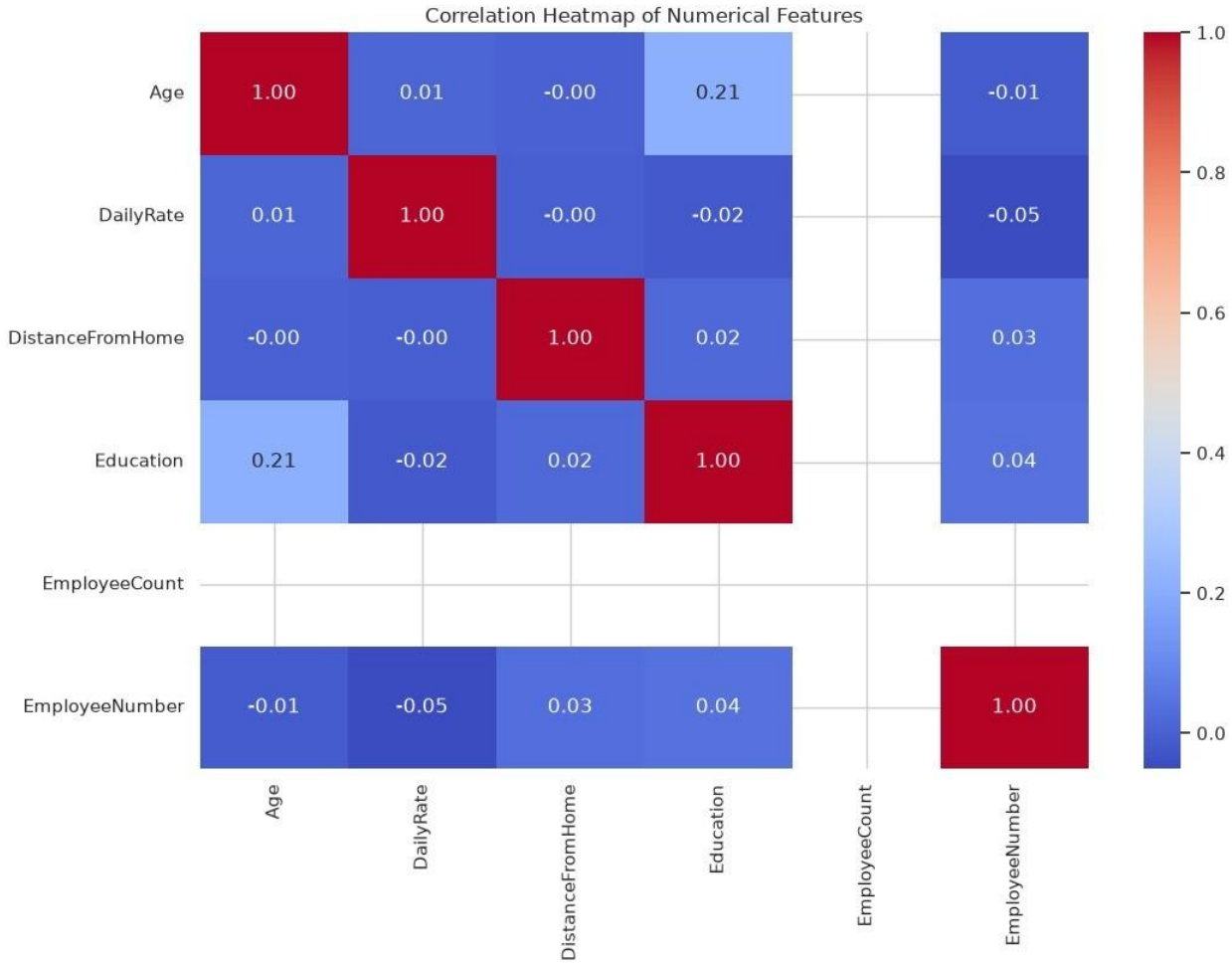



```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(12, 8))

sns.heatmap(
numeric_df.corr(),
annot=True,
cmap="coolwarm",
fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```

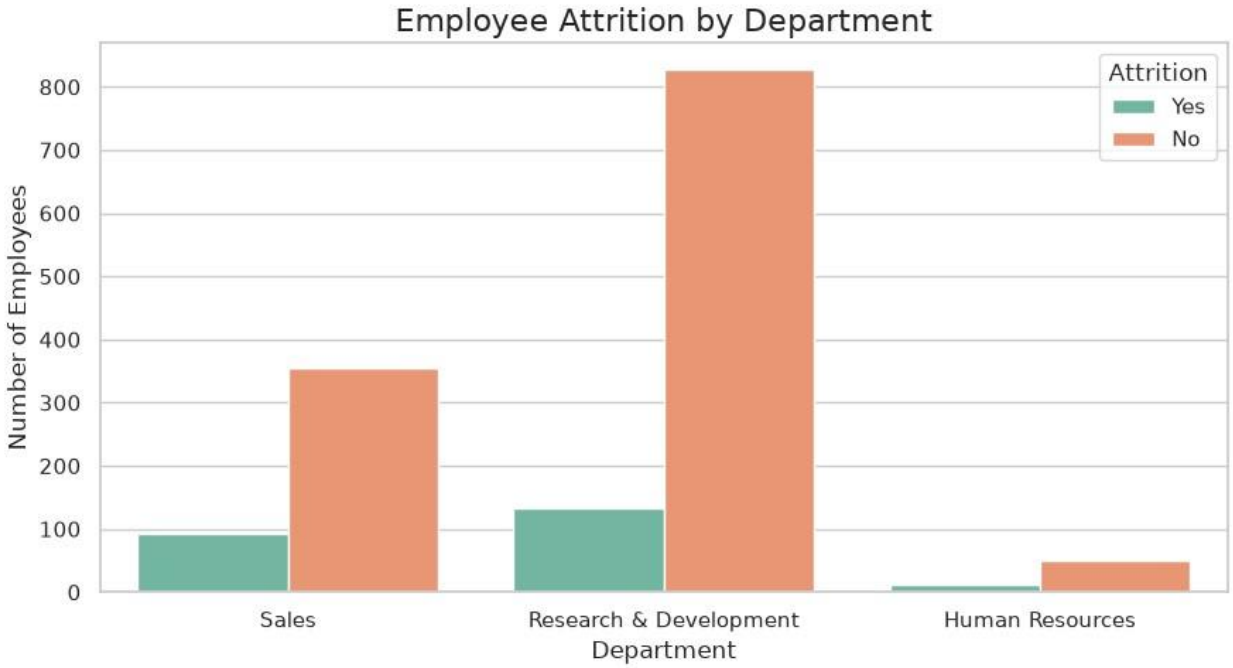



```
plt.figure(figsize=(9, 5))

sns.countplot(
data=df,
x="Department",
hue="Attrition",
palette="Set2"
)

plt.title("Employee Attrition by Department", fontsize=16)
plt.xlabel("Department", fontsize=12) plt.ylabel("Number
of Employees", fontsize=12) plt.legend(title="Attrition")

plt.tight_layout()
plt.show()
```


Employee Attrition by Department
Attrition
Yes
No
Number of Employees
0
100
200
300
400
500
600
700
800
Sales
Research & Development
Human Resources
Department

```
print("Total Employees:", len(df))

print("\nDepartment-wise Employees:")
print(df["Department"].value_counts())

print("\nAttrition Distribution:")
print(df["Attrition"].value_counts())
print("\nAverage Age:", df["Age"].mean())

print("\nAverage Daily Rate:", df["DailyRate"].mean())
```

```
Total Employees: 1470

Department-wise Employees:
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64

Attrition Distribution:
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Average Age: 36.923809523809524

Average Daily Rate: 802.4857142857143
```

QUESTIONS AND ANSWERS

**1. Why is data visualization essential in data analytics?**

Data visualization presents data through charts and graphs, making complex information easier to understand. It helps identify trends, patterns, comparisons, and outliers and supports better decision-making.

**2. Difference between Matplotlib and Seaborn**

- Matplotlib: General-purpose library with high customization.
- Seaborn: Built on Matplotlib and mainly designed for statistical visualization.

Seaborn is more suitable for statistical visualizations because it provides easy-to-use plots such as heatmaps, box plots, and regression plots.

**3. Which chart is suitable for comparing sales across regions?**

A bar chart is most suitable because regions are categories, and bars make it easy to compare their sales values and identify the highest and lowest sales.

**4. What information do histogram and box plot provide?**

- Histogram: Shows the distribution and frequency of numerical data.
- Box plot: Shows median, quartiles, spread, and outliers.

**5. Purpose of a scatter plot**

A scatter plot shows the relationship between two numerical variables. It helps identify positive, negative, or no correlation and can also reveal clusters and outliers.

**6. What is a correlation heatmap?**

A correlation heatmap displays correlation values between variables using colors. It helps identify strongly correlated features, which can assist in removing redundant features during machine learning feature selection.

**7. Why is chart customization important?**

Titles, labels, legends, and colors make charts clear and easy to understand. Proper customization helps the viewer interpret the data correctly.

**8. Factors for selecting a visualization**

Consider:

- Type of data
- Purpose of analysis
- Number of variables
- Audience
- Data size
- Readability

Examples: line chart for trends, bar chart for comparison, histogram for distribution, scatter plot for relationships.

**9. How can misleading visualizations affect business decisions?**

Misleading charts can cause incorrect interpretation and poor decisions. For example, using a truncated Y-axis can make a small increase in sales appear very large, causing management to overestimate business growth.

**10. How does visualization support storytelling and business intelligence?**

Visualization converts raw data into clear and meaningful insights. Dashboards and charts help organizations monitor KPIs, identify trends, understand problems, and make quick and informed business decisions.